

EXPANSION SYSTEM

OVERVIEW

The Ground Area Expansion System handles the different areas the player can purchase and enables switching between them. It also handles unloading grid chunks when they're not being viewed. The primary script used by this system is `GroundAreaExpansion.cs`. Other scripts that are involved include `TileMeshLoadManager.cs`, `TileMeshLoader.cs`, and `CameraMove.cs`.

The `GroundAreaExpansion` class is the primary manager of the entire system. The `TileMeshLoadManager` class unloads the meshes of all the tiles on chunks that aren't being viewed, and the `CameraMove` class moves the camera to new chunks when they're being viewed.

Whenever switch-chunk arrows (the little blue arrows on the left or right side of the screen) are clicked, they will call either the `MoveLeft()` or `MoveRight()` function of the `GroundAreaManager` class. That initiates the ground chunk switching process.

At the beginning of every chunk switch, the `GroundAreaExpansion` class tells the tile mesh load manager to load in the chunk that is about to be viewed. To do this, the tile mesh load manager gets all the tiles on the next grid chunk using the `GridChunk` class's `GetAllTilesOnGridChunk()` function. Then it grabs the `TileMeshLoader` class on each of the tiles and uses their `LoadTileMesh()` function, which turns on the tile's mesh. After the next chunk is loaded, the camera is told to begin moving to the new chunk using `CameraMove`'s function `MoveCamera()`. Once the camera has fully moved over to the new chunk, the old chunk is unloaded by the tile mesh loader, using the `LoadTileMesh` class's `UnloadTileMesh()` function. This process is easy to see if you view the game in the editor from the scene view while a chunk switch arrow is clicked.

The `Ground Area Expansion` class also creates new ground chunks whenever the player purchases the last one and the number of chunks is currently below the max number.

ASSOCIATED EVENTS

Many events (from the EventType class's enum "E") are associated with the ground chunk switching process. Here are some of the most important ones:

SwitchedCurrentGroundChunk

The ground chunk switching process just finished. In other words, the camera has finished fully moving into place.

SwitchedCurrentGroundChunkLate

Called immediately after the SwitchedCurrentGroundChunk event.

BeginSwitchingCurrentGroundChunk

The ground chunk switching process just began. In other words, the camera is just starting to move.

BeginSwitchingCurrentGroundChunkLate

Called immediately after the BeginSwitchingCurrentGroundChunkEvent

NewAreaUnlocked

Called when a new ground chunk is created.

PurchasedCurrentGroundChunk

The player just bought a ground chunk.

SwitchedChunkLeft

Same as SwitchedCurrentGroundChunk but is only called when the player switches to the left ground chunk.

SwitchedChunkRight

Same as SwitchedCurrentGroundChunk but is only called when the player switches to the right ground chunk.

BeginSwitchingChunkLeft

Same as BeginSwitchingCurrentGroundChunk but is only called when the player begins switching to the left ground chunk.

BeginSwitchingChunkRight

Same as BeginSwitchingCurrentGroundChunk but is only called when the player begins switching to the right ground chunk.

INTERNAL API

These are some important public functions from scripts in the Ground Chunk Expansion System.

GroundAreaExpansion.cs

void AddGroundChunk()

Creates a new ground chunk to the right of the current ones.

void MoveLeft()

Switches to the ground chunk to the left of the active ground chunk. Uses
SwitchToGroundChunk(ActiveGroundChunkNum - 1).

void MoveRight()

Switches to the ground chunk to the right of the active ground chunk. Uses
SwitchToGroundChunk(ActiveGroundChunkNum + 1).

void AddGroundChunk()

Adds a ground chunk. This is used when the player purchases a new area.

GameObject CreateNewVisibleGroundForChunk(int groundChunkNumber)

Instantiates a new patch of dirt for each area. Used when loading and unloading chunks. The game object it returns is the ground object it instantiated.

TileMeshLoadManager.cs

void UnloadGridChunk(int gridChunkIndex)

Unloads all the meshes in the chunk corresponding to gridChunkIndex.

void LoadGridChunk(int gridChunkIndex)

Loads all the meshes in the grid chunk corresponding to gridChunkIndex.

CameraMove.cs

void MoveCamera(Vector3 _moveDestination, float cameraMoveTime)

Moves the camera from its current location to the input _moveDestination in "cameraMoveTime" seconds.